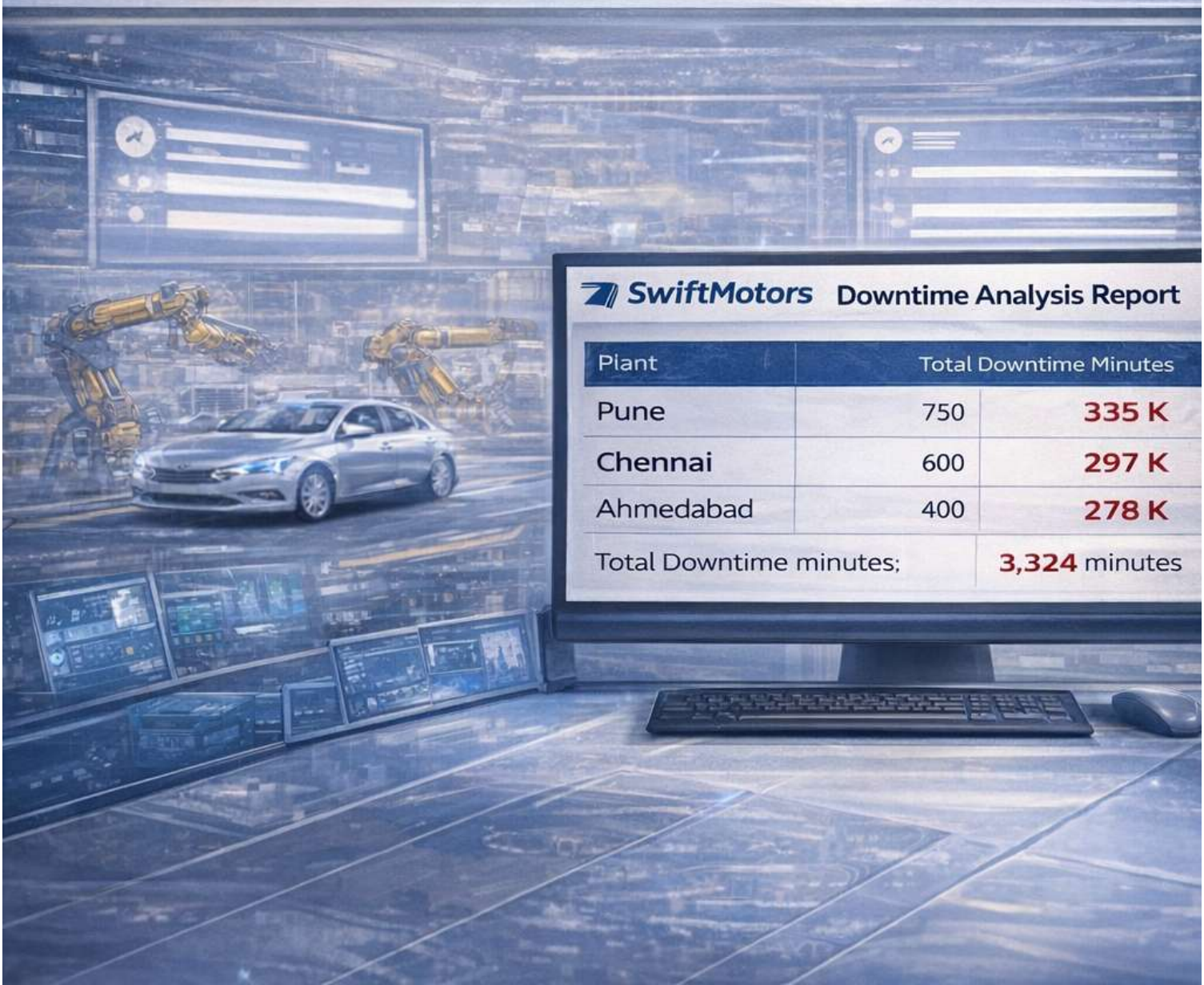
CASE STUDY QUESTION

Q3 [Defect Type Distribution Report

Using **Report Visualize Prompt**, create a visualization where Quality Managers can select chart type to view Defect Type Distribution across all plants.

- Add a second prompt for **Date Range**.
- Summarize the **top 3 most frequent defect types** as a bullet list text object within the report.
- Apply font color **Navy Blue** for all column headers.



SwiftMotors Downtime Analysis Report

Plant		Total Downtime Minutes
Pune	750	**335 K**
Chennai	600	**297 K**
Ahmedabad	400	**278 K**
Total Downtime minutes;		**3,324** minutes

CASE STUDY
Defect Type Distribution Reporting

Using Report Visualize Prompt, create a report for Quality Managers to view defect type distribution across all plants. Allow selection of chart type, define a date range filter, and summarize the top defect types. Apply Navy Blue font color for all column headers.

Defect Type Distribution Report

Chart Type: Bar Chart

Date Range: 2024-01-01 to 2024-04-30

Run

01 Set Up Report Layout

Design the Defect Type Distribution report allowing for: - Chart Type selection (e.g, Bar Chart, Pie Chart)

- Date Range selection to filter the report data

02 Group Data & Add Prompts

Configure prompts for:

- **Defect Type** (e.g, Scratch, Misalignment, Burr)
- Date Range (e.g. calendar picker for custom date ranges).

03 Calculate Defect Rate

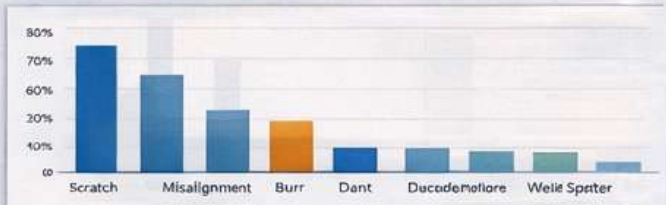
Calculate the Defect Rate using the formula (Defective Units / Total Units Produced) * 100.

- Group data by Defect Type to visualize defect distribution across all plants.

Defective / Produced *106

Defect Type: Bar Chart | 2024-01-01 to 2024-04-30 | Run

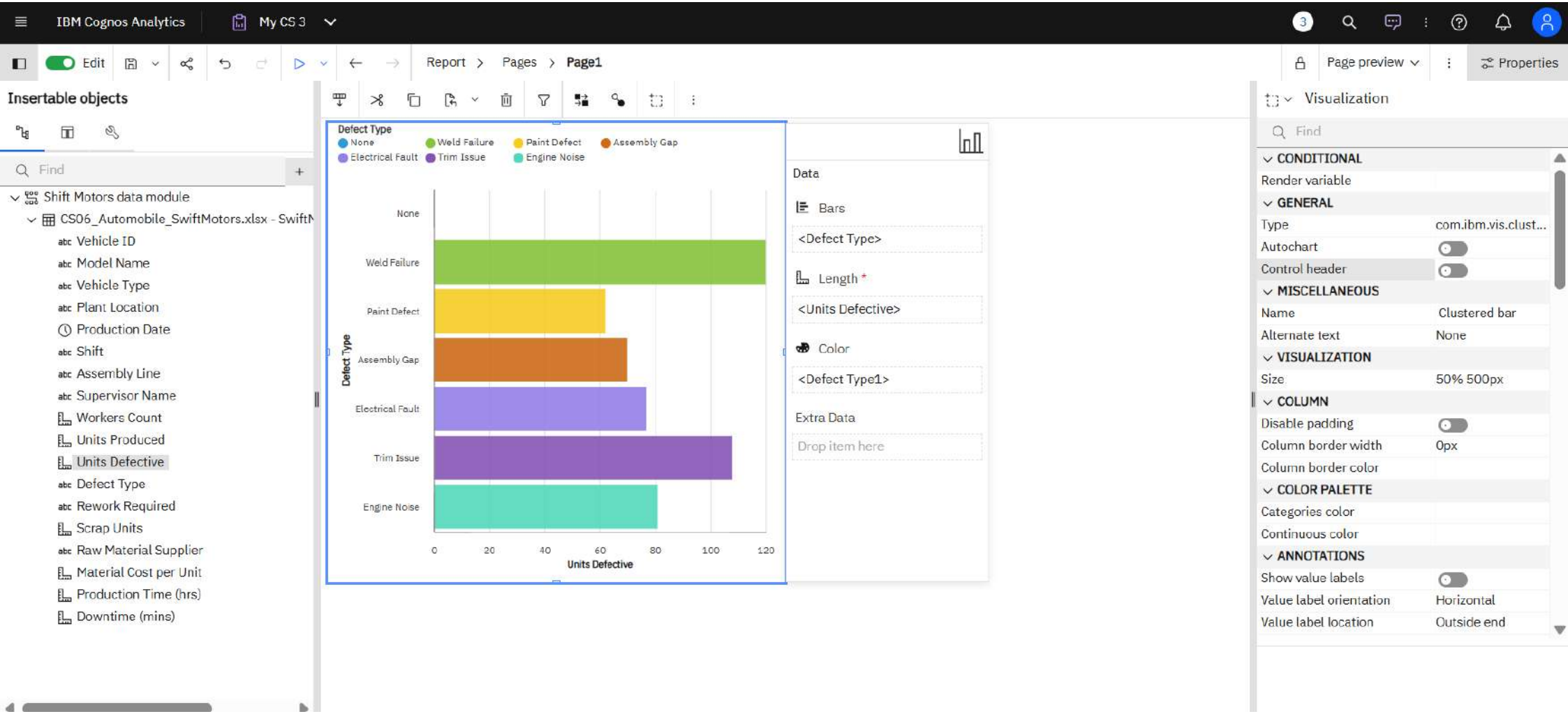
Defect Type Distribution Report



Shart	Total Units	Disfactive Units	Defect Rate	Defect Rate
Scratch	8,32	8,35	3.9%	6,9%
Misalignment	202	358	3.8%	3.2%
Burr	440	25%	2.3%	3.7%
Dent	202	1.3%	1.9%	1.9%
Discoloration	302	1.3%	1.2%	1.9%
Weld Spatter	315	1.3%	1.1%	1.1%

Top 2 Defect Types

- **Scratch** 352
- **Misalignment** 267
- **Burr** 195


IBM Cognos Analytics
My CS 3
Edit
Report > Pages > Page1
Page preview
Properties
Insertable objects
Find
Shift Motors data module
CS06_Automobile_SwiftMotors.xlsx - SwiftM
Vehicle ID
Model Name
Vehicle Type
Plant Location
Production Date
Shift
Assembly Line
Supervisor Name
Workers Count
Units Produced
Units Defective
Defect Type
Rework Required
Scrap Units
Raw Material Supplier
Material Cost per Unit
Production Time (hrs)
Downtime (mins)
Defect Type
None
Weld Failure
Paint Defect
Assembly Gap
Electrical Fault
Trim Issue
Engine Noise
Defect Type
0
20
40
60
80
100
120
Units Defective
Data
Bars
<Defect Type>
Length *
<Units Defective>
Color
<Defect Type1>
Extra Data
Drop item here
Visualization
Find
CONDITIONAL
Render variable
GENERAL
Type
com.ibm.vis.clust...
Autochart
Control header
MISCELLANEOUS
Name
Clustered bar
Alternate text
None
VISUALIZATION
Size
50% 500px
COLUMN
Disable padding
Column border width
0px
Column border color
COLOR PALETTE
Categories color
Continuous color
ANNOTATIONS
Show value labels
Value label orientation
Horizontal
Value label location
Outside end

IBM Cognos Analytics | My CS 3

Edit | Report > Prompt pages > **Prompt page1** | Page preview | Properties

Insert data objects

Find

- Shift Motors data module
  - CS06_Automobile_SwiftMotors.xlsx - SwiftM
    - Vehicle ID
    - Model Name
    - Vehicle Type
    - Plant Location
    - Production Date
    - Shift
    - Assembly Line
    - Supervisor Name
    - Workers Count
    - Units Produced
    - Units Defective
    - Defect Type
    - Rework Required
    - Scrap Units
    - Raw Material Supplier
    - Material Cost per Unit
    - Production Time (hrs)
    - Downtime (mins)

Double-click to edit text

DATE IN RANGE

Defect Type	Units Defective
Assembly Gap	70
Electrical Fault	77
Paint Defect	62

Double-click to edit text

Cancel | < Back | Next > | Finish

No properties

Select an object to see its properties

ap2.ca.analytics.ibm.com/bi/?perspective=authoring&closeWindowOnLastView=true

Maintenance: The upgrade is now complete. Click on More Info to see what actions may be necessary and to subscribe to future events

Dismiss More info

IBM Cognos Analytics My CS 3

HTML

Mode: Interactive View Filters

DATE IN RANGE

From:

Parameter1

Lowest value

To:

Parameter1

Highest value

Defect Type	Units Defective
Assembly Gap	70
Electrical Fault	77
Paint Defect	62

Cancel < Back Next > Finish

ap2.ca.analytics.ibm.com/bi/?perspective=authoring&closeWindowOnLastView=true

Maintenance: The upgrade is now complete. Click on More Info to see what actions may be necessary and to subscribe to future events

Dismiss More info

IBM Cognos Analytics My CS 3

HTML

Mode: Interactive View Filters

DATE IN RANGE

From:

Parameter1

Parameter1

Jan 2, 2024

Jan 3, 2024

Jan 4, 2024

Jan 5, 2024

Jan 9, 2024

Jan 12, 2024

Jan 13, 2024

Jan 14, 2024

Jan 15, 2024

Jan 19, 2024

Jan 21, 2024

Jan 22, 2024

Jan 23, 2024

Jan 27, 2024

Jan 28, 2024

Jan 31, 2024

Defect Type	Units Defective
Assembly Gap	70
Electrical Fault	77
Paint Defect	62

Cancel < Back Next > Finish

ap2.ca.analytics.ibm.com/bi/?perspective=authoring&id=i9940579791D74EA1A0299FFF69DCB4C5&objRef=i9940579791D74EA1A0299FFF69DCB...

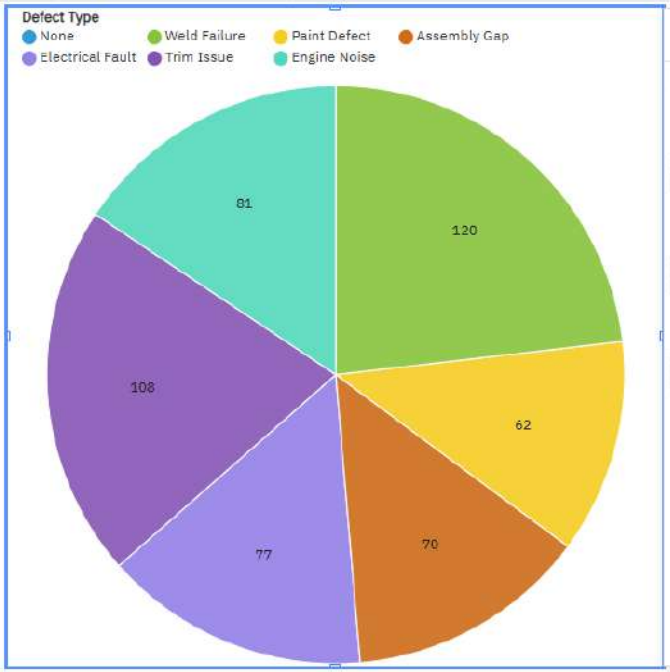
IBM Cognos Analytics | * My CS 3

Edit | Report > Pages > Page1 | Page preview | Properties

Insertable objects

Find

- Shift Motors data module
  - CS06_Automobile_SwiftMotors.xlsx - SwiftM
    - Vehicle ID
    - Model Name
    - Vehicle Type
    - Plant Location
    - Production Date
    - Shift
    - Assembly Line
    - Supervisor Name
    - Workers Count
    - Units Produced
    - Units Defective
    - Defect Type
    - Rework Required
    - Scrap Units
    - Raw Material Supplier
    - Material Cost per Unit
    - Production Time (hrs)
    - Downtime (mins)



Data

Segments *

<Defect Type>

Size *

<Units Defective>

Extra Data

Drop item here

Visualization

Find

CONDITIONAL	
Render variable	
GENERAL	
Type	com.ibm.vis.pie
Autochart	
Control header	
MISCELLANEOUS	
Name	Clustered bar
Alternate text	None
VISUALIZATION	
Size	50% 500px
PIE	
Donut hole radius	0
Pie border width	1px
Pie border color	
Sort order	None
COLOR PALETTE	
Segments color	
ANNOTATIONS	
Display %	
Show value labels	
Item label format	Value

ap2.ca.analytics.ibm.com/bi/?perspective=authoring&id=i9940579791D74EA1A0299FFF69DCB4C5&objRef=i9940579791D74EA1A0299FFF69DCB...

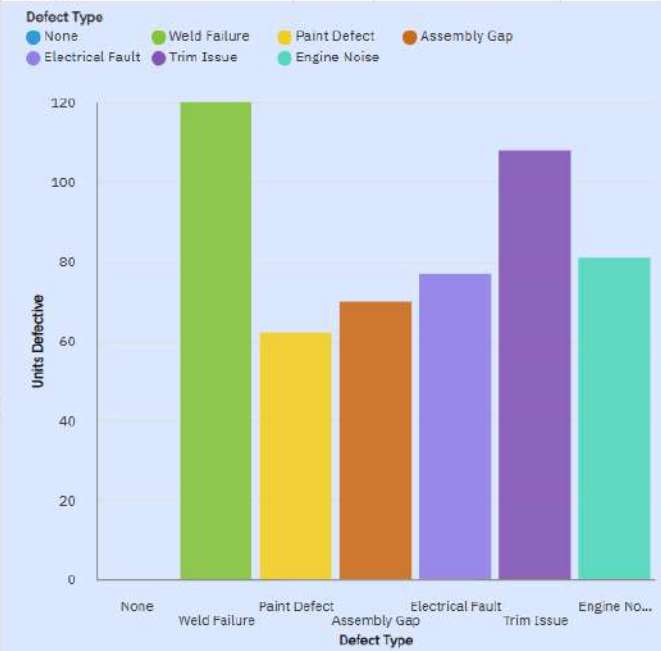
IBM Cognos Analytics | * My CS 3

Edit | Report > Pages > Page1 | Page preview | Properties

Insertable objects

Find

- Shift Motors data module
  - CS06_Automobile_SwiftMotors.xlsx - SwiftM
    - Vehicle ID
    - Model Name
    - Vehicle Type
    - Plant Location
    - Production Date
    - Shift
    - Assembly Line
    - Supervisor Name
    - Workers Count
    - Units Produced
    - Units Defective
    - Defect Type
    - Rework Required
    - Scrap Units
    - Raw Material Supplier
    - Material Cost per Unit
    - Production Time (hrs)
    - Downtime (mins)



Page body

Find

- CONDITIONAL
  - Conditional styles
  - Style variable
- BOX
  - Border
  - Padding
  - Box type
- COLOR & BACKGROUND
  - Background image
  - Background effects
  - Background color
  - Foreground color
- FONT & TEXT
  - Font
  - Horizontal alignment
  - Vertical alignment
  - White space
  - Spacing & breaking
  - Direction & justification
- POSITIONING
  - Size & overflow
- MISCELLANEOUS